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Duhem, Quine and the Other Dogma

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Abstract: A resemblance between positions held by Duhem and Quine has led to the conjunction of their names: one speaks of 'Duhem-Quine.' Quine's doctrine is expressed in 'Two Dogmas of Empiricism'; it was by refuting the second that he wound up in the company of Duhem. But there is also the first, the analytic-synthetic distinction. Quine questions it too, but one can wonder whether Duhem does: over and over he emphasizes the troublesome synthetic character of physics by contrasting it with the rigid necessity of mathematics—in which analytic truths are generally held to figure paradigmatically. Indeed his discussion of crucial experiments seems to rest on the very distinction Quine questions; on an adherence to the first of the two dogmas. But what about the second? An individual experiment cannot be crucial; Duhem also uses the term 'experiment,' however, for a class of equivalent but very different experiments, which all measure the same thing. I propose a structuralist set-theoretical characterization of an 'abstract experiment' or 'experimental structure,' whose (physical) models are the individual equivalent experiments; and consider the status of the structure, which may have an almost analytic 'cruciality,' while referring to the world through its physical models.